Felipe Vaz Peres

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EXPERIENCE

Huna **Remote**
Data Scientist *Sep 2025 - Present*

- Develop interpretable risk-stratification models for early cancer detection, prioritizing coefficient-level transparency so clinicians can audit each prediction's reasoning
- Co-authored abstract accepted at ASCO 2026 on PSA-based prostate cancer risk stratification across 2,978 men from 7 Brazilian diagnostic laboratory networks. Developed a model achieving 0.77 AUC and 0.98 negative predictive value, reducing false-positive classifications by 35.1% relative to standard PSA cutoff screening

Center for Microbial Secondary Metabolites, Technical University of Denmark **Remote**
Guest Researcher *Jan 2024 - Jan 2025*

- Developed a metagenomic pipeline processing 100+ marine samples, generating 40 high-quality genomes (>90% completeness) from long-read sequencing data
- Applied ML-based contig classification to distinguish prokaryotic/eukaryotic sequences, enabling unbiased genome assembly across diverse microbial communities
- Standardized biosynthetic gene cluster annotation across 40 genomes, cataloging 148 full-length BGCs and establishing consistent nomenclature
- Reduced manual execution by developing end-to-end pipeline with benchmarking and documentation; presented findings at lab meetings

Computational, Evolutionary and Systems Biology Laboratory, University of São Paulo **Piracicaba, SP**
Graduate Researcher *Jan 2022 - Jan 2025*

- Applied ML models (SVM, Random Forest, CNN) identifying 8.4M ncRNAs across 48 sugarcane genotypes, establishing the largest pan-RNAome reference to date
- Analyzed co-expression networks (1M+ nodes) discovering 7,000+ ncRNA-protein modules, revealing functional roles across diverse genetic backgrounds
- Functionally annotated 250K+ ncRNAs establishing a framework for investigating biological functions in this complex polyploid crop
- Contributed to 5 open-source projects across bacteria, microalgae, and plants, building automated Nextflow pipelines used by 3 research groups

Genera **Remote**
Bioinformatics Intern *Jan 2020 - Mar 2020*

- Enhanced human ancestry calculator analyzing 650K+ SNPs, improving resolution for 72 population groups including underrepresented South American populations
- Developed documentation for Genera's ancestry calculator

Computational, Evolutionary and Systems Biology Laboratory, University of São Paulo **Piracicaba, SP**
Undergraduate Researcher *May 2019 - May 2021*

- Processed 3.7B reads (1.1TB+) across HPC infrastructure, constructing the sugarcane pan-transcriptome (16M transcripts) covering 48 genotypes
- Developed automated Snakemake pipeline assembling 48 transcriptomes, enabling systematic comparison of 600K+ protein variants, defining core/accessory/exclusive protein sets
- Presented reproducible pipeline design at ISMB, demonstrating scalability across polyploid crop species

EDUCATION

University of São Paulo, Center of Nuclear Energy in Agriculture **Piracicaba, SP**
M.Sc. in Bioinformatics *Jan 2025*

- Genomics and Bioinformatics, From Gene to Trait, Mobile Genetic Elements, Bioinformatics and Microbiomes

Federal University of São Carlos, Center of Agrarian Sciences **Araras, SP**
B.S. in Biotechnology *Jan 2022*

- Statistics, Bioinformatics, Molecular Biology, Microbial Genetics, Breeding and Genetics, Genetic Engineering

COMPUTATIONAL SKILLS

Programming: Python, Bash, R
Machine Learning: scikit-learn, model validation, interpretable modeling, MLflow
Cloud & Infrastructure: Databricks, GCP, Unix/Linux, HPC (SGE, PBS, LSF, SLURM), git
Bioinformatics & Genomics: transcriptomics, genomics, metagenomics, NGS (short and long reads), gene expression, co-expression networks, functional annotation, Snakemake, Nextflow

HONORS AND AWARDS

- **3rd place**: Liga Brasileira de Bioinformática - The largest bioinformatics competition in Latin America — *Jun 2025*
- **Graduate Research Fellowship:** National Council for Scientific and Technological Development — *April 2022*
- **3rd place:** Developed an automated variant calling pipeline in under 48 hours at *Mendelics Challenge* — *Nov 2021*
- **Honorable mention:** International Symposium of Undergraduate Research at University of São Paulo — *Oct 2020*
- **Undergraduate Research Fellowship:** São Paulo Research Foundation — *April 2019*
- **2nd place:** Synthetic Biology Hackathon at University of São Carlos — *Aug 2018*

SOFTWARE

Additional work can be seen on my GitHub profile: https://github.com/felipevzps

Selected work

- **KAPT** - Nextflow pipeline for proteome inference and annotation through de novo transcriptome assembly
- **R2C** - Nextflow pipeline to build (gene) co-expression networks
- **sugarcane_RNAome** - Multi-genotype analyses of long-ncRNAs in sugarcane
- **ContFree-NGS** **-** Open source software that removes sequences from contaminating organisms in NGS datasets
- **YAATAP** - Snakemake pipeline for downloading raw RNA-seq datasets and performing transcriptome assemblies
- **seabed-symphony** - Pipeline for BGCs identification in microbial community of marine sediments

SELECTED PRESENTATIONS

2022 **Python: From zero to your first software**
Presented at workshop - Python Workshop for Biological Data, Online Conference - UNICAMP

An introduction to programming for bioscientists
Presented at workshop - Genetics and Molecular Biology Meeting, Online Conference - UNICAMP

2021 **Removing reads from contaminating organisms in NGS datasets**
Poster - Brazilian Symposium on Bioinformatics, Online Conference - Federal University of Minas Gerais

2020 **Analysis of *de novo* transcriptome assemblies in sugarcane**
Selected Talk (Top 15%) - International Symposium of Undergraduate Research, University of São Paulo